



DYNAMIC STOPPING DISTANCE: SENSE-TO-ACTUATION DELAY AS PHYSICAL DISPLACEMENT

Translating Milliseconds of Computational Latency into Physical Centimeters of Unguided Collision Hazard

$$d_{\text{stop}}(v_0, \Delta t_{\text{delay}}) = \underbrace{v_0 \cdot \Delta t_{\text{delay}}}_{\text{Phase 1: Linear Reaction Travel (Coasting during computational delay)}} + \underbrace{\frac{v_0^2}{2a_{\text{max}}}}_{\text{Phase 2: Quadratic Braking Distance (Mechanical deceleration to rest)}}$$

🚧 Physical Clearance Barrier ($d_{\text{clearance}} = 35 \text{ cm}$)

✓ **Nominal Case (P_{50}):** $v_0 = 1.0 \text{ m/s}$, $\Delta t = 30 \text{ ms}$, $a_{\text{max}} = 2.0 \text{ m/s}^2$



⚠️ **Tail Latency Spike (P_{99}):** $v_0 = 1.0 \text{ m/s}$, $\Delta t = 230 \text{ ms}$, $a_{\text{max}} = 2.0 \text{ m/s}^2$

